RoHS Compliance Status

Stackpole Electronics has joined the worldwide effort to reduce the amount of lead in electronic components and to meet the various regulatory requirements now prevalent, such as the European Union's directive regarding "Restrictions on Hazardous Substances" (RoHS 3). As part of this ongoing program, we periodically update this document with the status regarding the availability of our compliant components. All our standard part numbers are compliant to EU Directive 2011/65/EU of the European Parliament as amended by Directive (EU) 2015/863/EU as regards the list of restricted substances.

Resistors

Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
ASR	Anti-Surge Resistor	Axial	YES	99.3/0.7 Sn/Cu
ASRM	Mini Anti-Surge Resistor	Axial	YES	99.3/0.7 Sn/Cu
ASRM14	Mini Anti-Surge Resistor	Axial	YES(1)	99.3/0.7 Sn/Cu
BR	Bare Element Current Sensing Resistor	Special	YES	100% Matte Sn
BVM	Ceramic Housed Vertical Bracket Mount Resistor	Special	YES	100% Matte Sn
CB	Ceramic Housed with Axial Leads Power Resistor (Standard)	Axial	YES	100% Matte Sn
CBF	Ceramic Housed with Axial Leads Power Resistor (with standoff)	Axial	YES	100% Matte Sn
CBT	High Power Ceramic House Wirewound with Tabs	Axial	YES	100% Matte Sn
CD	Axial Leaded Zero Ohm Resistor	Axial	YES	100% Matte Sn
CDM	Axial Leaded Zero Ohm Resistor (Mini)	Axial	YES	100% Matte Sn
CF	Carbon Film Resistor	Axial	YES	100% Matte Sn
CFF	Carbon Film Resistor (Flameproof)	Axial	YES	100% Matte Sn
CFFM	Carbon Film Resistor (Flameproof - mini)	Axial	YES	100% Matte Sn
CFM	Carbon Film Resistor (Mini)	Axial	YES	100% Matte Sn
CFQ	Carbon Film Resistor (Tin Plating on Copper Wire)	Axial	YES	100% Matte Sn
CFQM	Carbon Film Resistor (Tin Plating Mini on Copper Wire)	Axial	YES	100% Matte Sn
CRNM	Non-Magnetic Audio Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Cu
CSFA	Automotive Grade High Current Jumper	SMD	YES	100% Matte Sn over Ni
CSM	Molded Metal Alloy Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSNL	Metal Alloy Current Sensing Chip Resistor	SMD	YES	100% Matte Sn over Ni
CSR	Thick Film Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSRF	Foil on Ceramic Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSRN	Thick Film Current Sensing Resistor (Narrow)	SMD	YES	100% Matte Sn over Ni
CSRT	Thin Film Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSRW	Thick Film Long Side Termination Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSS	Ultra-Precision Metal Alloy Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSSH	Ultra-Precision Metal Alloy Current Sensing Resistor (High Power)	SMD	YES	100% Matte Sn over Ni
CSSK	Kelvin Termination Metal Alloy Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
CSSU	High Power Metal Alloy Current Sensing Resistor	SMD	YES	100% Matte Sn over Ni
ESD	Low Capacitance ESD Suppressor	SMD	YES	100% Matte Sn over Ni
ESDU	Ultra-Low Capacitance ESD Suppressor	SMD	YES	100% Matte Sn over Ni
EWT	Non-Flammable Edgewound Tubular Wirewound Resistor	Special	YES	100% Matte Sn
FCR	Trimmable Thick Film Resistor	SMD	YES(1)	100% Matte Sn over Ni
FRC	Fusible Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
FRN	Axial Leaded Fusing Resistor	Axial	YES	99.3/0.7 Sn/Cu
HCC	High Current Chassis Mount Shunt Resistor - Terminal	SMD	YES	100% Copper
	Sizes 8218, 8518, and 8236 Sensing Terminal	SMD	YES	Cu (Ag plating/Over 3 µm)
HCJ	High Current Jumper Chip Resistor	SMD	YES	100% Matte Sn over Ni
HCS	High Current Metal Alloy Shunt Resistor	SMD	YES	100% Copper
HCSK	Kelvin Termination Metal Alloy Shunt Resistor	SMD	YES	100% Copper
HCSM	High Power Metal Alloy Current Shunt Resistor	SMD	YES	100% Matte Sn over Ni
HDM	Moisture Resistant Carbon Film Resistor	Axial	YES	100% Matte Sn
HGC	Thick Film Precision High Resistance Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
HMC	High Resistance Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni

Note (1): RoHS Compliant by means of exemption 7c-I.

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RoHS Compliance Status

Resistors				
Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
HVA	High Voltage Axial Leaded Resistor	Axial	YES(1)	100% Matte Sn
HVAM	Precision High Voltage Leaded Resistor ***DISCONTINUED*** (January 19, 2022)	Axial	YES	100% Matte Sn
HVC	High Voltage Thick Film Resistor	SMD	YES(1)	100% Matte Sn over Ni
HVCR	High Voltage Low VCR (100% VCR Screened) Chip Resistor	SMD	YES(1)	100% Matte Sn ("T")
HVCD	High Voltage Chip Divider	SMD	YES(1)	100 Matte Sn
HVD	High Voltage Plate Resistor Divider	Radial Special	YES(1)	96.3/3.7 Sn/Ag
HVR	High Voltage Radial Leaded Plate Resistor	Radial Special	YES(1)	100% Matte Sn
JW	Tin Plated Jumper Wire Resistor	Axial	YES	100% Matte Sn
KAL	Aluminum Housed Chassis Mount Resistor (Standard)	Special	YES	100% Matte Sn
KAW	Aluminum Housed WW Resistor Insulated wire Leads and Crimped Connectors	Special	YES	100% Matte Sn
LCB	Ceramic Housed Current Sensing Resistor with 2 Leads (Standard)	Axial	YES	100% Matte Sn
LCBF	Ceramic Housed Current Sensing Resistor with 2 Leads (with Standoff)	Axial	YES	100% Matte Sn
LVM	Ceramic Housed Vertical Mount Low Resistance Resistor (Ribbon Element)	Radial	YES	100% Matte Sn
MCB	Ceramic Housed General Purpose Metal Oxide Element Resistor (standard)	Axial	YES	100% Matte Sn
MCBF	Ceramic Housed General Purpose Metal Oxide Element Resistor (with standoff)	Axial	YES	100% Matte Sn
MCJ	Metal Element High current Jumper Chip Resistor	SMD	YES	100% Matte Sn over Ni
MG	High Voltage Metal Glaze Resistor (Standard)	Axial	YES(1)	100% Matte Sn
MGE	High Voltage Metal Glaze Resistor (Epoxy Coating)	Axial	YES(1)	100% Matte Sn
MGM	High Voltage Metal Glaze Leaded Resistor (Mini)	Axial	YES(1)	100% Matte Sn
MGME	High Voltage Metal Glaze Leaded Resistor (Mini - Epoxy Coating)	Axial	YES(1)	100% Matte Sn
MHL	Metal Clad Wirewound Resistor	Special	YES	100% Matte Sn
MLF	Metal Film Melf Resistor (Standard)	SMD	YES	100% Matte Sn over Ni
MLFA	Metal Film Melf Resistor (AEC-Q200 Qualified)	SMD	YES	100% Matte Sn over Ni
MLFM	Metal Film Melf Resistor (Mini)	SMD	YES	100% Matte Sn over Ni
MPR	Flameproof Metal Plate Resistor	Radial Special	YES	100% Matte Sn
MR	Low Resistance Value Leaded Resistor - Molded 2 Leads	Axial Kelvin	YES	100% Matte Sn
MRS	Low Resistance Value Leaded Resistor (High Stability)	Axial Kelvin	YES	100% Matte Sn
MVM	Ceramic Housed Vertical Mount Wirewound Resistor (Metal Oxide)	Radial	YES	100% Matte Sn
MWW	General Purpose and Precision Wirewound Resistor (Molded)	Axial	YES	100% Matte Sn
NCB	Welded Ceramic Housed Axial Leaded Wirewound Resistor (Non-Inductive)	Axial	YES	100% Matte Sn
NKAL	Aluminum Housed Chassis Mount Resistor (Non-Inductive)	Special	YES	100% Matte Sn
NKAW	Aluminum Housed WW Resistor Insulated wire Leads and Crimped Connectors	Special	YES	100% Matte Sn
NMWW	General Purpose and Precision Wirewound Resistor (Non-Inductive, Molded)	Axial	YES	100% Matte Sn

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Resistors				
Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
NSM	Surface Mount Wirewound Resistor (Non-Inductive)	SMD	YES	100% Matte Sn
NSZ	Ceramic Housed Wirewound Resistor with Specialty Leads	Radial	YES	99.3/0.7 Sn/Cu
NVM	Ceramic Housed Vertical Mount Wirewound Resistor (Non-Inductive)	Radial	YES	100% Matte Sn
NWW	General Purpose and Precision Wirewound Resistor (Non-Inductive)	Axial	YES	100% Matte Sn
NWWP	Pulse Withstanding Wirewound Resistor (Non-Inductive)	Axial	YES	100% Matte Sn
NWWS	General Purpose and Precision Wirewound Resistor (Non-Inductive, Mini)	Axial	YES	100% Matte Sn
PCB	Ceramic Housed - Commercial Grade	Radial	YES	100% Matte Sn
PCF	Carbon Film Resistor (Panasert CF14)	Axial	YES	100% Matte Sn
PCFM	Carbon Film Resistor (Panasert CFM12)	Axial	YES	100% Matte Sn
PCFQ	Carbon Film Resistor (Tin Plating on Copper Wire - Panasert)	Axial	YES	100% Matte Sn
PRNF	General Purpose Metal Film Resistor (Panasert)	Axial	YES	99.3/0.7 Sn/Cu 100% Matte Sn
RACF	Thick Film Surface Mount Chip Resistor Array Concave Terminations ***DISCONTINUED*** *(Nov. 15, 2019)*	SMD	YES(1)	100% Matte Sn over Ni
RAF	Flat Termination chip Resistor Array	SMD	YES(1)	100% Matte Sn over Ni
RAVF	Convex Termination Thick Film Chip Resistor Array	SMD	YES(1)	100% Matte Sn over Ni
RAVS	Convex Anti-Sulfur Chip Resistor Array	SMD	YES(1)	100% Matte Sn over Ni
RC	Carbon Composition Leaded Resistor - ***DISCONTINUED*** (Mar. 31, 2023)	Axial	YES	100% Matte Sn
RGC	Precision Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RHC	High Power Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMAN	Thick Film High Power Aluminum Nitride Substrate Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMCA	Automotive Grade Anti-Sulfur and AEC Qualified Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMCF	General Purpose Thick Film Chip Resistor (Standard Power)	SMD	YES(1)	100% Matte Sn over Ni
RMCG	Gold Barrier Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMCP	General Purpose High Power Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMCS	Sulfur Resistant Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMCW	Wide Termination Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMEA	Automotive Grade Anti-Sulfur Chip Resistor	SMD	YES	100% Matte Sn over Ni
RMEF	General Purpose Thick Film Chip Resistor 100% Lead Free RoHS Compliant Without Exemption	SMD	YES	100% Matte Sn over Ni
RMHT	Thick Film Automotive Grade High Temperature Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RMWA	Automotive Grade Wide Terminal Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RNAN	Thin Film Precision High Power Aluminum Nitride Substrate Chip Resistor	SMD	YES	100% Matte SN over Ni
RNCA	Automotive Grade Anti-Sulfur Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCF	Precision Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCL	Thin Film High Power Anti-Surge Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCE	High Temperature High Stability Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCP	High Power Anti-Sulfur Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCQ	High Frequency Thin Film Chip Resistors	SMD	YES	100% Matte Sn over Ni
RNCS	Moisture Resistant Precision Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCS-AS	Anti-Corrosive Anti-Sulfur Tantalum Nitride Replacement Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCU	Ultra-High Power Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCV	Automotive Grade High Voltage Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNCW	Thin Film Wire-Bondable Chip Resistor - ***DISCONTINUED*** *(Jan. 17, 2018)*	SMD	YES	Gold Plating

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Resistors				
Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
RNF	General Purpose Metal Film Resistor (Standard)	Axial	YES	99.3/0.7 Sn/Cu 100% Matte Sn
RNHT	Automotive Grade High Temperature Thin Film Chip Resistor	SMD	YES	100% Matte Sn over Ni
RNMF	General Purpose Metal Film Resistor (Mini)	Axial	YES	99.3/0.7 Sn/Cu 100% Matte Sn
RNS	Ultra-Miniature Metal Film Resistor	Axial	YES	100% Matte Sn
RNV	High Voltage Anti-Moisture Metal Film Resistor	Axial	YES	100% Matte Sn over Ni
RNWA	Automotive Grade Wide Terminal Thin Film Precision Chip Resistor	SMD	YES	100% Matte Sn over Ni
RPC	Pulse Withstanding Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RPCA	Automotive Grade Surge Withstanding Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RPCQ	Automotive Grade High Power Pulse Withstanding Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RSF	General Purpose Metal Oxide Resistor	Axial	YES	99.3/0.7 Sn/Cu 100% Matte Sn
RSMF	Mini-Metal Oxide Resistor	Axial	YES	99.3/0.7 Sn/Cu 100% Matte Sn
RSPF	Flameproof Power Resistor	Axial	YES	99.3/0.7 Sn/Cu
RSPL	Flameproof Power Resistor Low Resistance (Low Values)	Axial	YES	99.3/0.7 Sn/Cu
RTAN	Tantalum Nitride Thin Film Chip Resistor	SMD	YES	100% Matte Sn
RVC	Medium Voltage Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RVCA	Automotive Grade High Voltage Thick Film Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
RVCU	High Voltage Low VCR Chip Resistor	SMD	YES(1)	100% Sn over Ni
RWT	Thermal Fusing Vertical Mount Wirewound Resistor	Radial	YES	100% Matte Sn
SM	Surface Mount Wirewound Resistor (Standard)	SMD	YES	100% Matte Sn
SMH	Surface Mount Wirewound Resistor (High Power)	SMD	YES	100% Matte Sn
SP	Fusing Wirewound Resistor	Axial	YES	100% Matte Sn
SPR	Discharge Path Resistor	Axial	YES	99.3/0.7 Sn/Cu
SPRM	Discharge Path Resistor (Mini)	Axial	YES	99.3/0.7 Sn/Cu
TCB	Ceramic Housed Current Sensing Resistor with 4 Leads	Special (4 Leads)	YES	100% Matte Sn
TMJ	Thermal SMD Jumper	SMD	YES	100% Matte Sn over Ni
TMR	Low Resistance Value Resistor (Molded 4 Leads)	Axial Kelvin	YES	100% Matte Sn
TNC	Thick Film NTC Thermistor	SMD	YES	100% Matte Sn over Ni
TR	TO-220 and TO-247 Style Power Resistor	Radial Special	YES(1)	100% Matte Sn
UHV	Ultra-High Voltage Chip Resistor	SMD	YES(1)	100% Matte Sn over Ni
VM	Ceramic Housed Vertical Mount Resistor (Standard WW)	Radial	YES	100% Matte Sn
WCB	Welded Ceramic Housed Axial Leaded Wirewound Resistor (Standard)	Axial	YES	100% Matte Sn
WCBF	Welded Ceramic Housed Axial Leaded Wirewound Resistor (With Standoff)	Axial	Yes	100% Matte Sn
WRC	Conformal Coated Welded Wirewound Resistor	Axial	YES	100% Matte Sn
WVM	Ceramic Housed Vertical Mount Wirewound Resistor (Precision Wirewound)	Radial	YES	100% Matte Sn
WW	General Purpose and Precision Wirewound Resistor (Standard)	Axial	YES	100% Matte Sn
WWF	General Purpose Fusing Wirewound Resistor	Axial	YES	100% Matte Sn
WWP	Pulse Withstanding Wirewound Resistor	Axial	YES	100% Matte Sn
WWS	General Purpose and Precision Wirewound Resistor (Mini)	Axial	YES	100% Matte Sn
WWV	High Voltage Pulse Withstanding Wirewound Resistor	Axial	YES	100% Matte Sn

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RoHS Compliance Status

Capacitors				
Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
CML	Multilayer Ceramic Chip Capacitor	SMD	YES	100% Matte Sn over Ni

Varistors				
Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
AV	Automotive SMD Varistor (Standard)	SMD	YES	100% Matte Sn over Ni
AVY	Automotive SMD Varistor (High Temperature)	SMD	YES	100% Matte Sn over Ni
CV	Low and Medium Voltage Leaded Varistor	Leaded	YES	100% Matte Sn
CVH	Medium Voltage Leaded Varistor	Leaded	YES	100% Matte Sn
DV	Low and Medium Voltage SMD Varistor	SMD	YES	Ag/Pd
PV	Low and Medium Voltage Plastic Encapsulated SMD Varistor	SMD	YES(1)	100% Matte Sn
RV	Standard and High energy Metal Oxide Varistor	Leaded	YES	96.5/3/0.5 Sn/Ag/Cu
SV	Special Medium Voltage Leaded Varistor	Leaded	YES	100% Matte Sn
ZV	Low Voltage SMD Varistor (Standard)	SMD	YES(1)	100% Matte Sn over Ni
ZVY	Low Voltage SMD Varistor (High Temperature)	SMD	YES(1)	100% Matte Sn over Ni
ZVE	ESD Suppression SMD Varistor	SMD	YES(1)	100% Matte Sn over Ni
ZVL	Low Voltage Leaded Varistor (Standard)	Leaded	YES	100% Matte Sn
ZVYL	Low Voltage Leaded Varistor (High Temperature)	Leaded	YES	100% Matte Sn
ZVX	Low Energy, Low Capacitance SMD Varistor	SMD	YES	100% Matte Sn over Ni

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Inductors				
Standard Product Series	Description	Package / Termination Type	Standard Series RoHS Compliant	Lead-Free Termination Composition
FB	Multilayer Ferrite Chip Bead Inductor	SMD	YES	100% Matte Sn
LDRS	SMD Shielded Power Inductor	SMD	YES	100% Matte Sn
LP	SMD Unshielded Power Inductor	SMD	YES	100% Matte Sn
LPC	SMD Unshielded Power Inductor	SMD	YES	100% Matte Sn
LPCS	SMD Shielded Power Inductor	SMD	YES	100% Matte Sn
LPM	Miniature SMD Power Inductor	SMD	YES	100% Matte Sn
LPS	SMD Shielded Power Inductor	SMD	YES	100% Matte Sn
LTF	Thin Film Chip Inductor	SMD	YES	100% Matte Sn over Ni
LWF	Ferrite Wirewound Inductor	SMD	YES	100% Matte Sn
LWW	Wirewound Ceramic Chip Inductor	SMD	YES	100% Matte Sn

Marisol Alvarado

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